CS & IT ENGINEERING

DIGITAL LOGIC

Chapter:- Logic Gates

Lecture No. 3



By- CHANDAN SIR

Recap of Previous Lecture



Topic

NOT, AND, OR

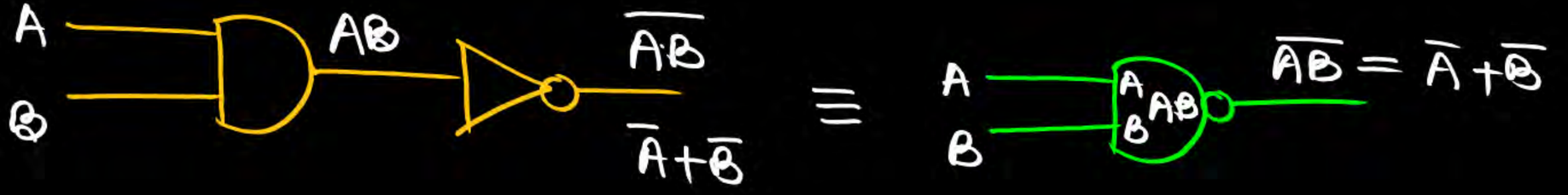
Topics to be Covered

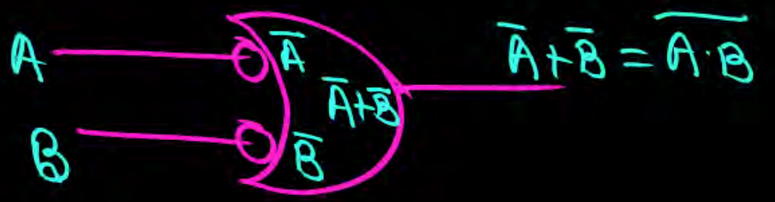


- NAND
- NOR GATE
- Discussion

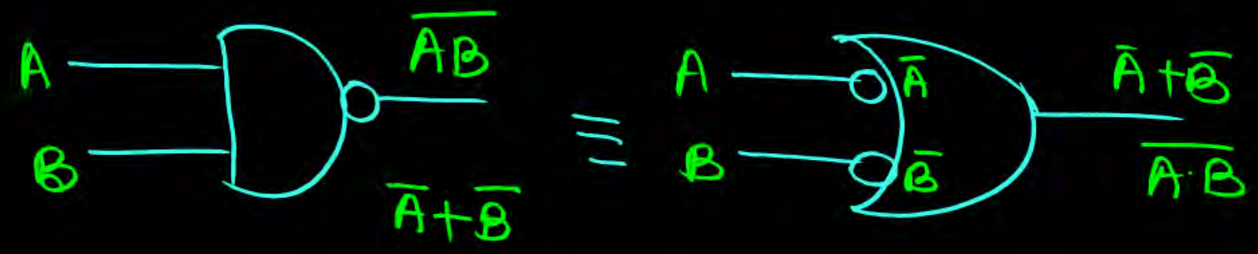


Topic : NAND & NOR GATE





Bubbled OR GATE ≡ NAND

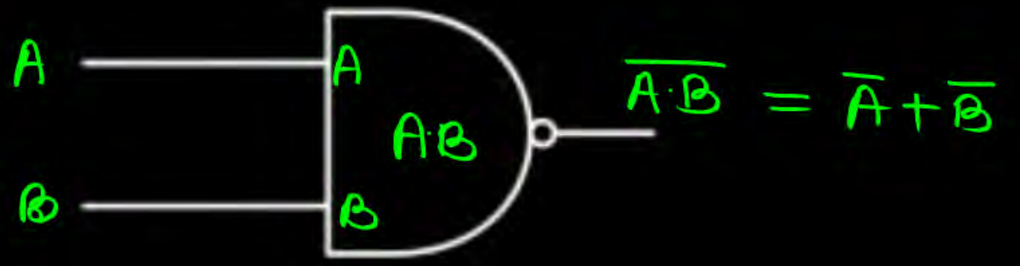

A
B
$\overline{AB}$
$\overline{A}+\overline{B}$
≡
A
B
$\overline{A}$
$\overline{B}$
$\overline{A}+\overline{B}$
$\overline{A\cdot B}$



Topic : NAND & NOR GATE

- **NAND GATE**

1. Symbol

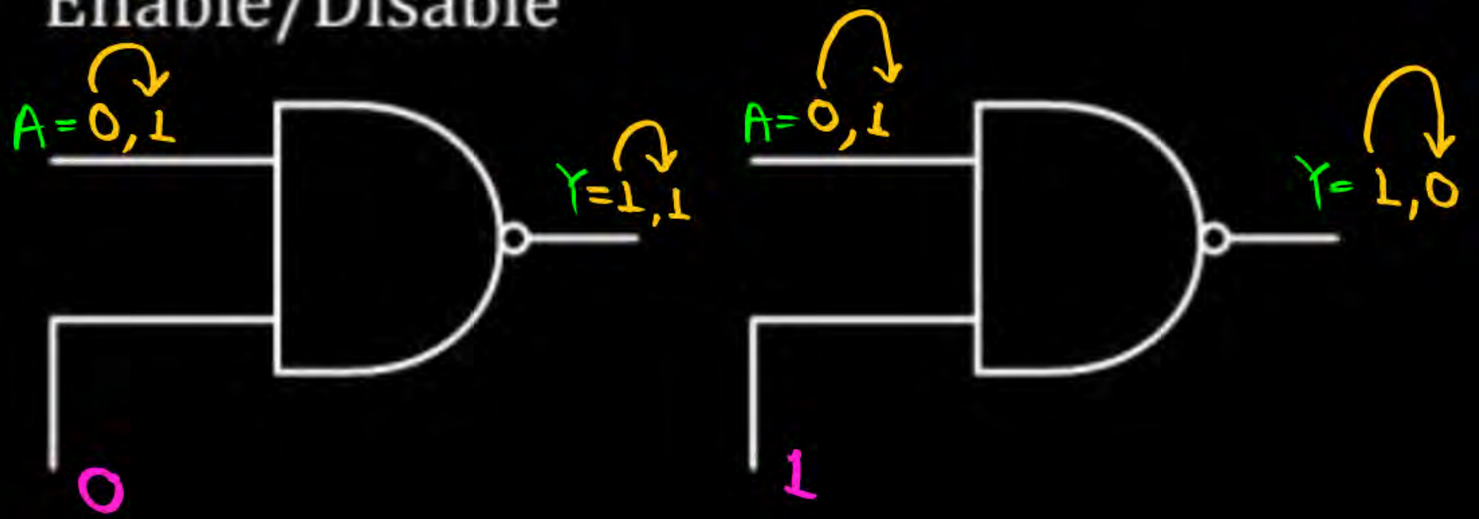


2. Truth Table

A	B	Y = $\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

- **NAND GATE**

3. Enable/Disable

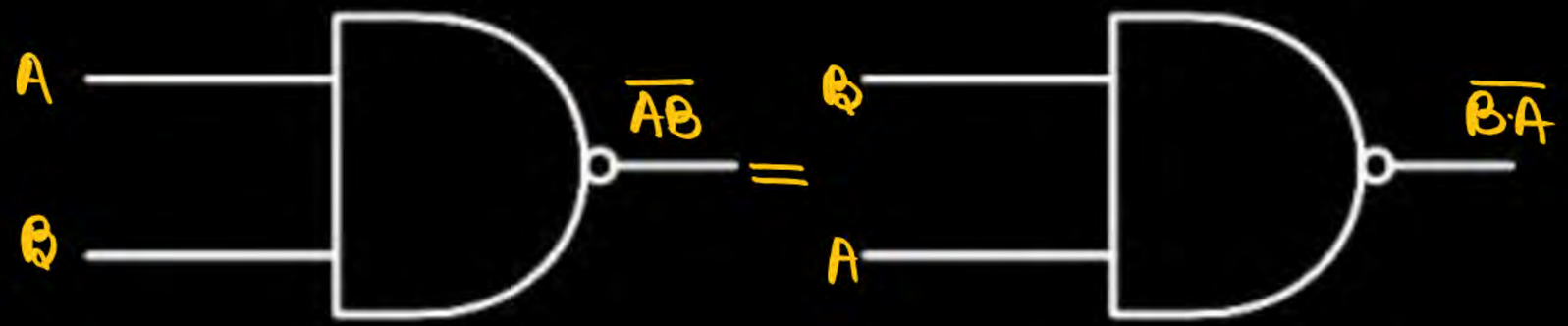


Control '0' Disable

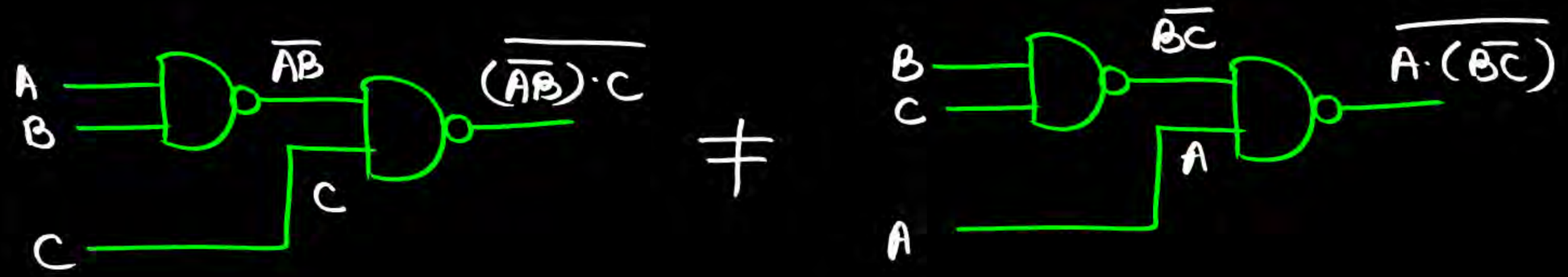
Control '1' Enable

- **NAND GATE**

4. Commutative Law

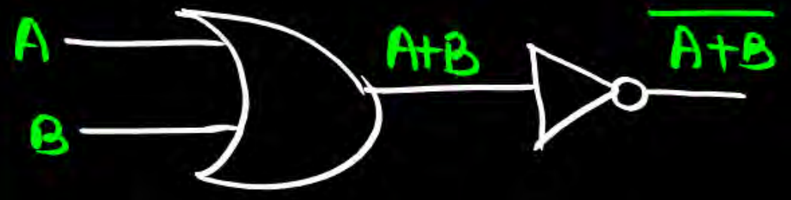


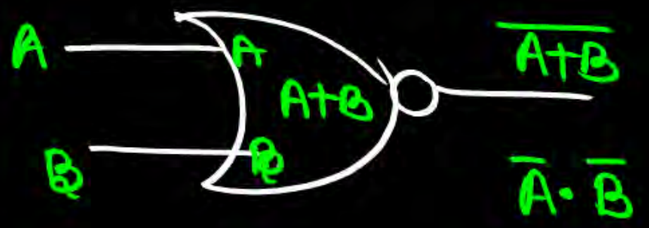
Associative Law :→

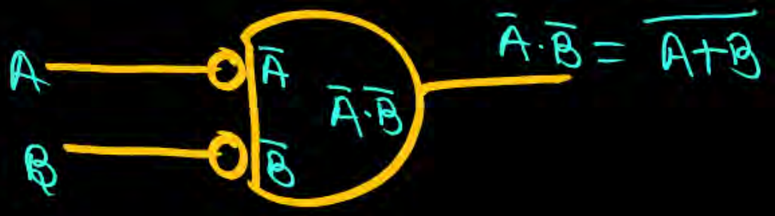


NAND does not follow associative law.

NOR GATE :→







Bubbled AND ≡ NOR

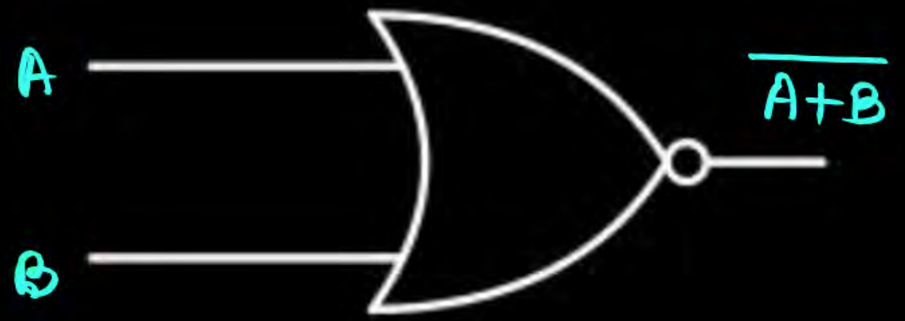




Topic : NAND & NOR GATE

- **NOR GATE**

1. Symbol

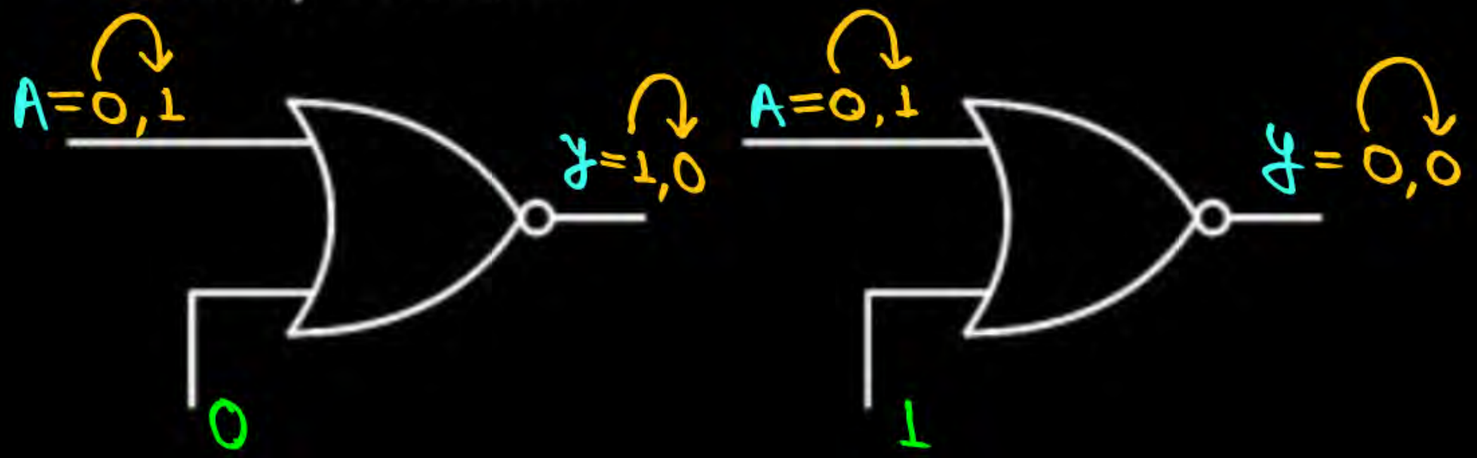


2. Truth Table

A	B	$Y=\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

- **NOR GATE**

3. Enable/Disable

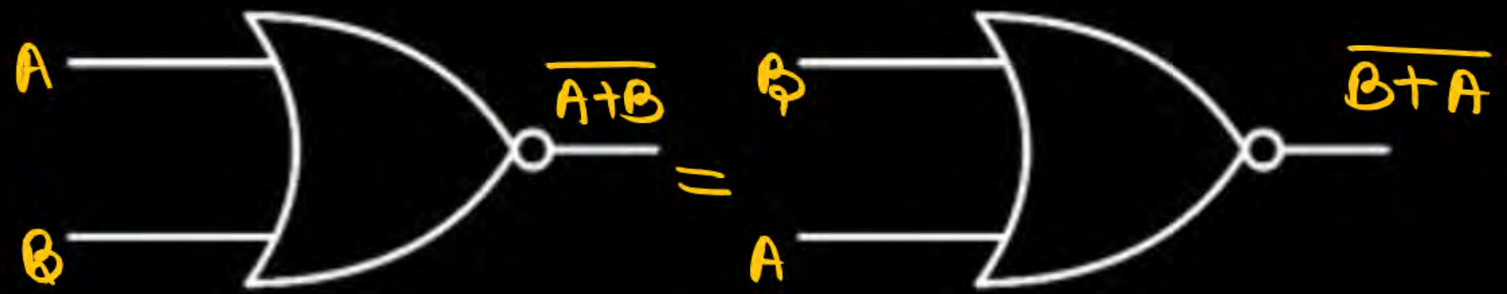


Control '0' Enable

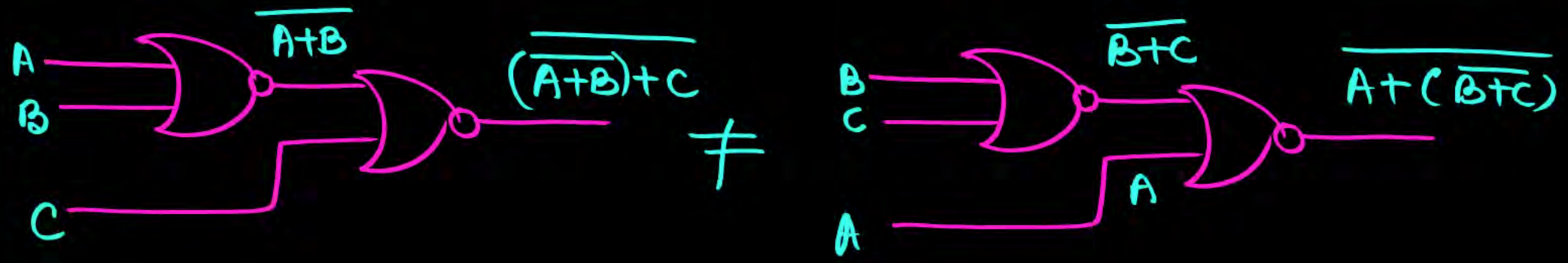
Control '1' Disable

- **NOR GATE**

4. Commutative Law ✓



Associative Law :→



Does not follow associative Law.

Eg NOTE

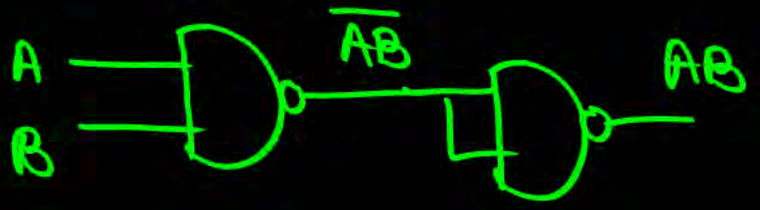
NAND, NOR are called universal logic

NAND AS A UNIVERSAL LOGIC

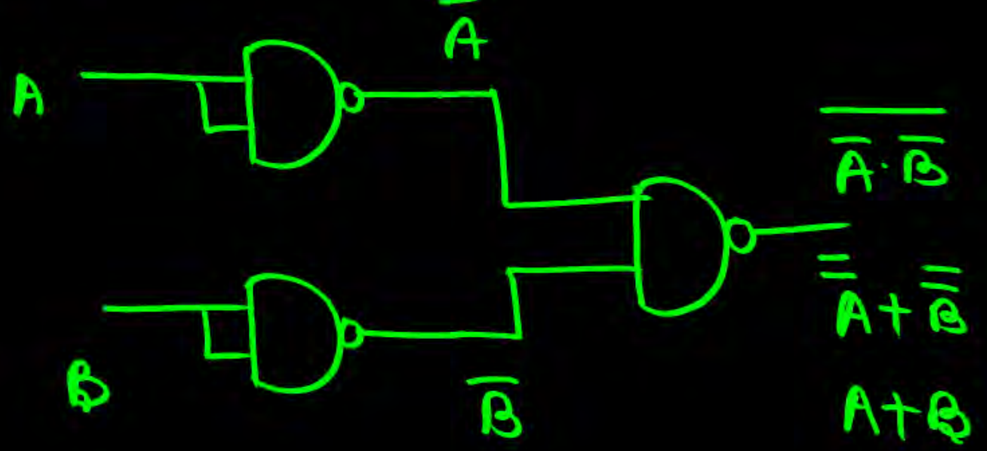
① NOT GATE :→



② AND GATE

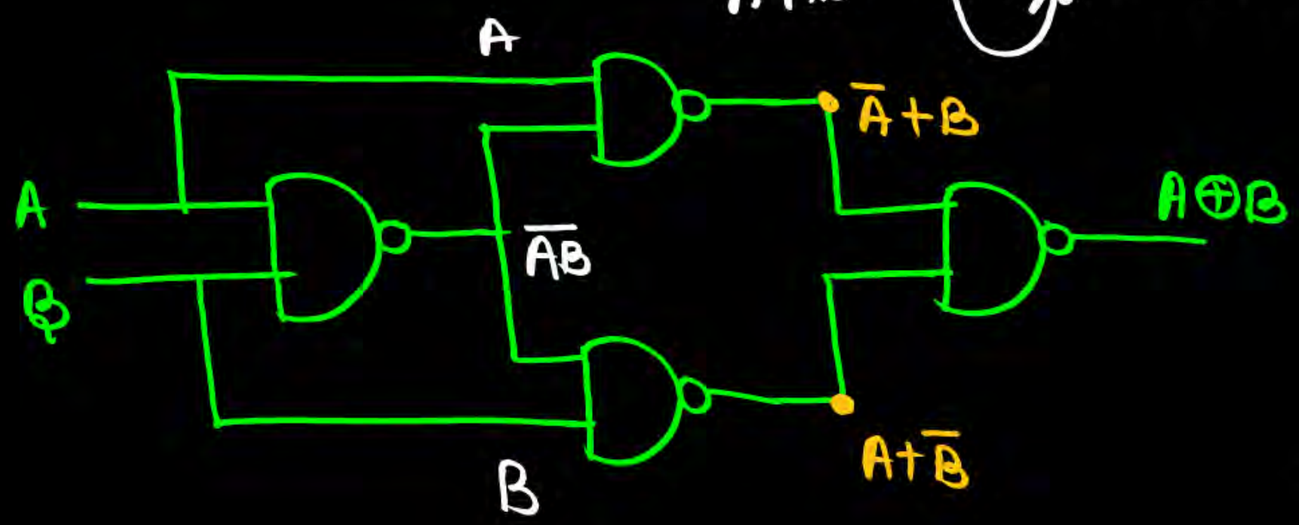


③ OR GATE :-



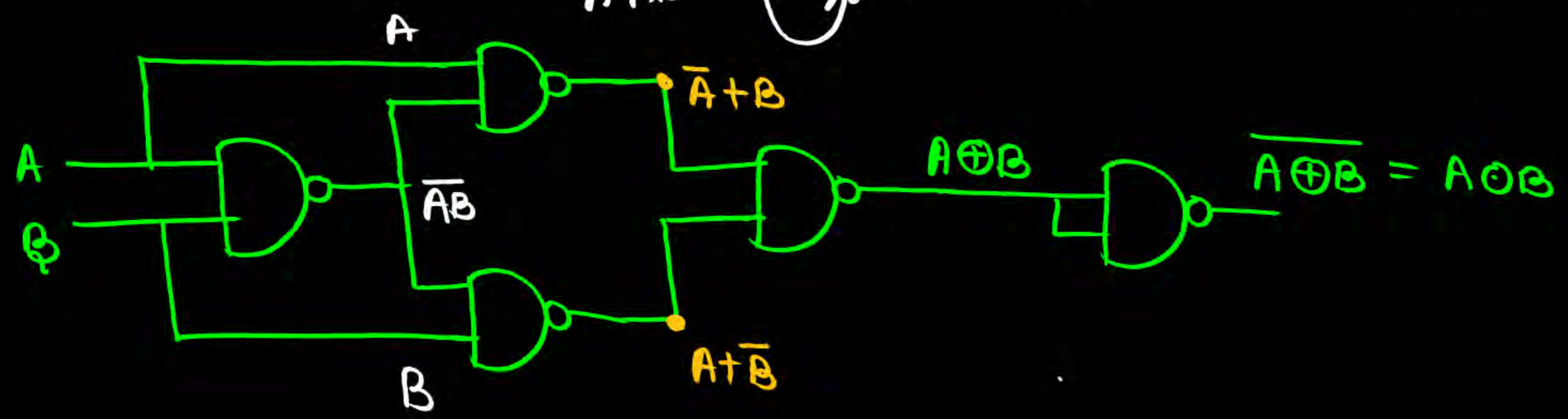
④ X-OR GATE :→ $A \oplus B = \bar{A}B + A\bar{B}$

$$\overline{A \cdot \bar{A}B} = \bar{A} + AB = (\bar{A}+A)(\bar{A}+B) = 1 \cdot (\bar{A}+B) = \bar{A}+B$$



⑤ X-NOR GATE $\Rightarrow A \odot B = \overline{A \oplus B}$

$$\overline{A \cdot \overline{AB}} = \bar{A} + AB = (\bar{A}+A)(\bar{A}+B) = 1 \cdot (\bar{A}+B) = \bar{A}+B$$



	NAND	NOR
NOT	1	1
AND	2	3
OR	3	2
X-OR	4	5
X-NOR	5	4
NAND	1	4
NOR	4	1

(MCQ)

#Q. Which of the following option is called universal logic?

A NAND

B NOR

C Both A & B

D None

(MCQ)

#Q. Which of the following option is called universal logic?

A NAND

B NOR

C AND

D OR

(MSQ)

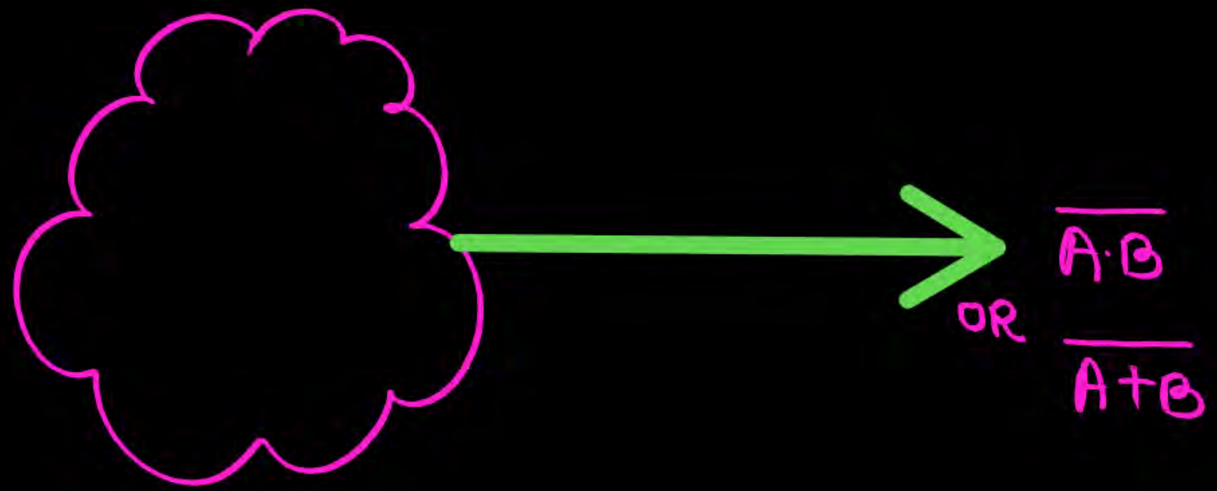
#Q. Which of the following option(s) is/are called universal logic?

A. $\left(\overline{A+B}\right)$

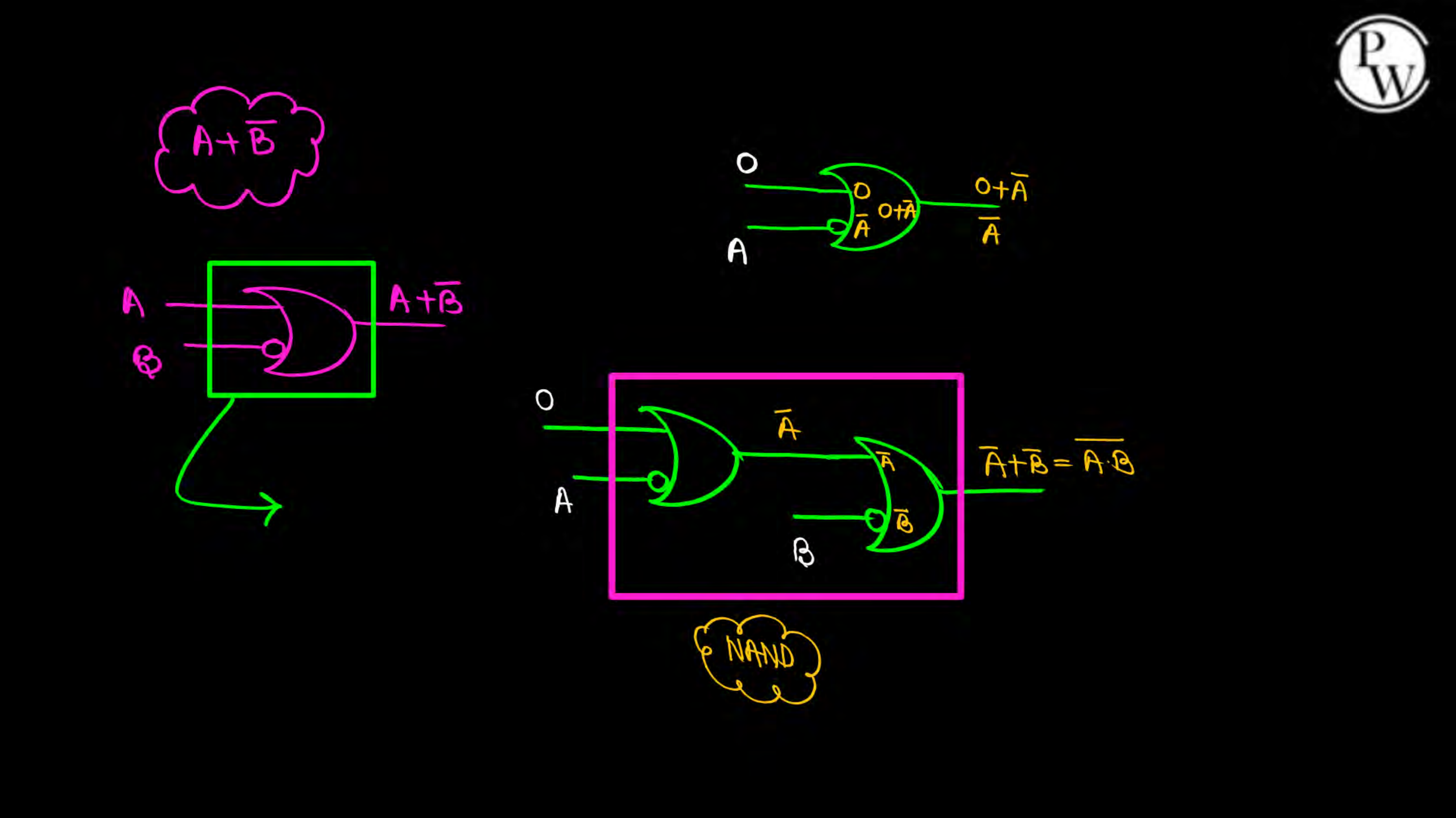
B. $\left(\overline{A \cdot B}\right)$

C. $\left(A+\overline{B}\right)$

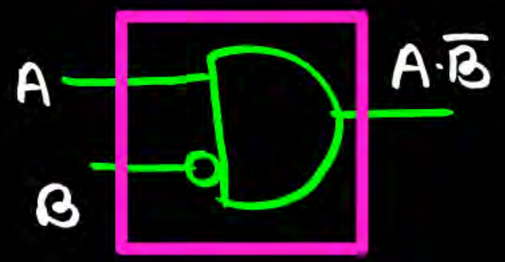
D. $A \cdot \overline{B}$

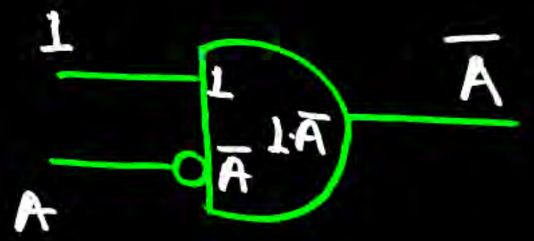


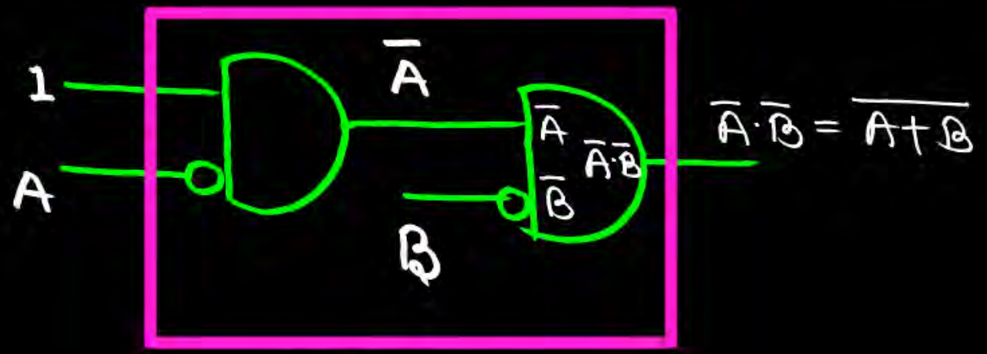
Universal


A+B̄
A
B
A+B̄
0
A
0
Ā
0+Ā
0+Ā
Ā
0
A
Ā
Ā
Ā+B̄ = A·B
B
B̄
B
NAND

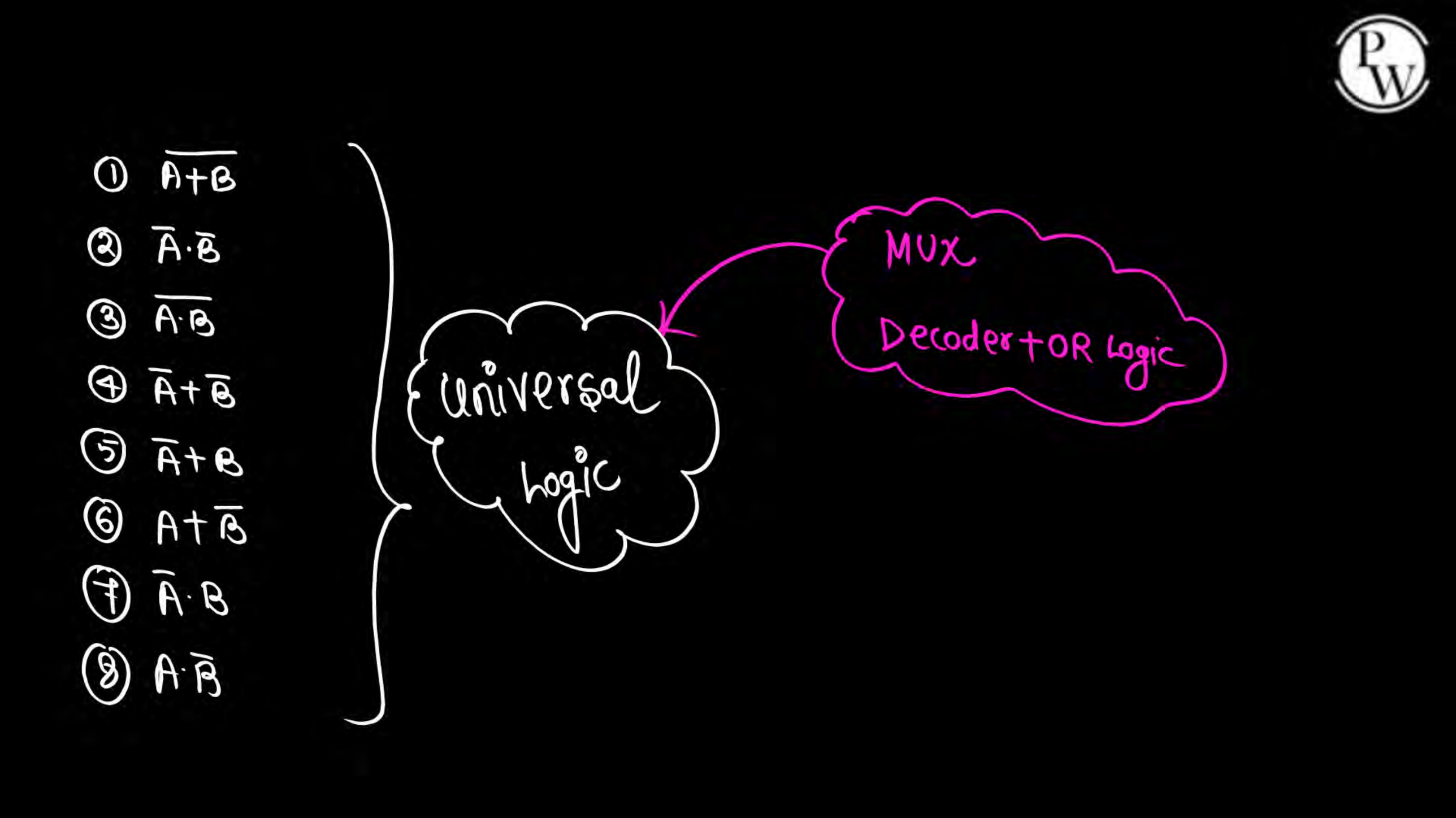
$A\cdot\bar{B}$







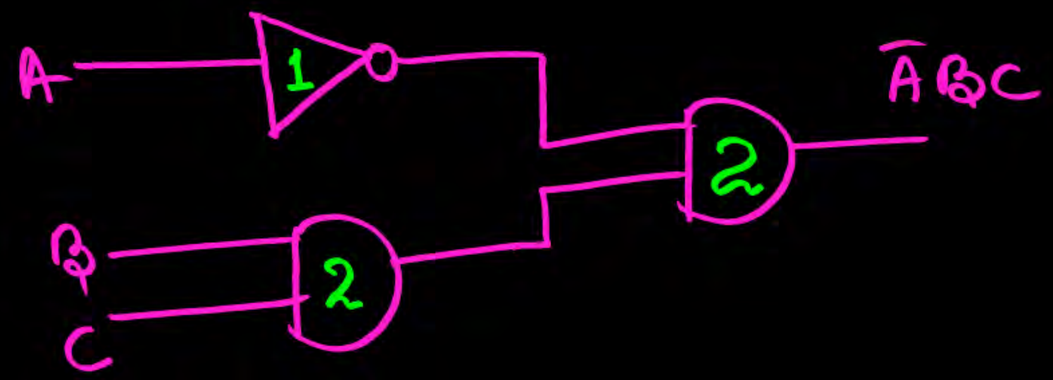
NOR


① $\overline{A+B}$
② $\bar{A}\cdot\bar{B}$
③ $\overline{A\cdot B}$
④ $\bar{A}+\bar{B}$
⑤ $\bar{A}+B$
⑥ $A+\bar{B}$
⑦ $\bar{A}\cdot B$
⑧ $A\cdot\bar{B}$
Universal Logic
MUX
Decoder + OR Logic

(MCQ)

#Q. Find the minimum number of two input NAND GATE required to implement the function-

$f(A, B, C) = \bar{A}BC$



$1+2+2=5$

A 3

B 5 ✓

C 6

D 4

(MCQ)

#Q. Find the minimum number of two input NAND GATE required to implement the function-

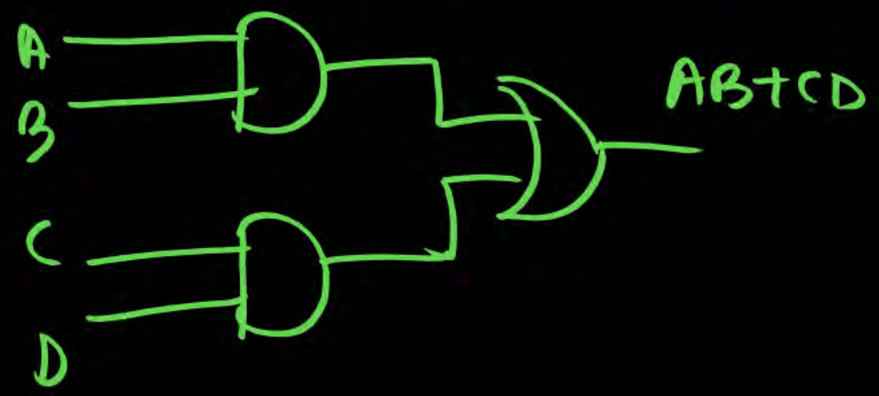
$f(A, B, C, D) = AB + CD$

A. 3

B. 5

C. 6

D. 7



#Q. Consider the following function f,

$f = A \oplus B \oplus \overline{AB}$

The number of 2 input NAND gate to implement the minimized function f is ____ .

#Q. The minimum number of two input NAND gates required to implement y = abcd is

A 3

B 4

C 5

D 6

#Q. Find the minimum number of two input NAND GATE required to implement the function-

$$f(A, B, C) = \bar{A}BC$$

A 3

B 6

C 5

D 4



2 Min Summary

- NAND
- NOR GATE
- Discussion

Thank You GW Soldiers!

